KERALA DIPLOMA CURRICULUM REVISION 2026

Course 6342C | 6 Credits: 4 | Open Elective | Source-grounded

# Big Data Concepts

 To familiarize with Big Data and its sources

OFFICIAL SOURCE DECLARATION

## How this lesson was prepared

The supplied official SITTTR syllabus PDF for Course 6342C is the authoritative source for course information, objectives, course outcomes, CO–PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

**Source file:** 6342C.pdf from the uploaded official SITTTR syllabus archive. **Course code and title:** preserved from the source.

COURSE DASHBOARD

## Official course information

**Course information**

| Item | Official information |
|---|---|
| Programme | Diploma in Artificial Intelligence & Machine Learning |
| Course code | 6342C |
| Course title | Big Data Concepts |
| Semester | 6 Credits: 4 |
| Course category | Open Elective |
| Periods per week | 4 (L:4 T:0 P:0) Periods per semester: 60 |
| Periods per semester | 60 |

| Item | Official information |
|---|---|
| Credits | 4 |
| Prerequisite | See the official syllabus PDF |
| Assessment | Use the official assessment section reproduced below |

### Modules

4 official module sections detected.

### Topic entries

18 source-aligned topic entries retained.

### Study mode

Theory and application emphasis.

STUDY METHOD

## How to use this lesson

### First pass

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

### Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

### Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

### Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

LEARNING OUTCOMES

# Objectives, outcomes and mapping

## Course objectives

-  To familiarize with Big Data and its sources
-  To understand the basics of data preprocessing and data visualization
-  To familiarize Big Data tools and techniques.

**Course outcomes**

| CO | Official outcome | Study level |
|---|---|---|
| CO1 | Familiarize the concepts of Big Data 12 Understanding | See official syllabus |
| CO2 | Outline the basic concepts in Big Data Analytics 15 Understanding Implementation of Exploratory Data Analysis using | See official syllabus |
| CO3 | 15 Applying Python Demonstrate Map Reduce programming in Hadoop | See official syllabus |
| CO4 | 16 Applying Environment | See official syllabus |

**CO–PO mapping evidence**

| Row | Extracted official mapping |
|---|---|
| CO1 | CO1 3 |
| CO2 | CO2 3 |
| CO3 | CO3 3 |
| CO4 | CO4 3 |

COVERAGE AUDIT

## Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

**Module coverage**

| Module | Official heading | Topic entries | Hours |
| --- | --- | --- | --- |
| Module 1 | Familiarize the concept of Big Data | 4 | As specified |
| Module 2 | Outline the basic concepts in Big Data Analytics | 4 | As specified |
| Module 3 | Implement of Exploratory Data Analysis using Python | 4 | As specified |
| Module 4 | Demonstrate Map Reduce programming in Hadoop Environment | 6 | As specified |

MODULE 1

# Familiarize the concept of Big Data

This module is presented from the official syllabus scope for Big Data Concepts. Use the topic cards for learning and the source transcript for exact coverage.

**As specified**
official hours

**CO-linked**
see outcomes

### Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

### Key facts

- Official module title: Familiarize the concept of Big Data
- Official duration: As specified
- Official topic entries captured: 4

- The full source excerpt is preserved below for coverage checking.

## Familiarize the concepts of Big Data.

Familiarize the concepts of Big Data. is an official topic in Module 1 of Big Data Concepts. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Familiarize the concepts of Big Data. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, familiarize the concepts of big data. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

**Topic study guide**

| Aspect | What to prepare |
|---|---|
| Definition/identity | What Familiarize the concepts of Big Data. means in the official course context |
| Study lens | Principle → components or stages → result → application |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 1-[illegible] Familiarize the concepts of Big Data. [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

## Describe Big data Analytics and its applications

Describe Big data Analytics and its applications is an official topic in Module 1 of Big Data Concepts. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Describe Big data Analytics and its applications using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, describe big data analytics and its applications should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

**Topic study guide**

| Aspect | What to prepare |
|---|---|
| Definition/identity | What Describe Big data Analytics and its applications means in the official course context |
| Study lens | Principle → components or stages → result → application |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 1-[illegible] Describe Big data Analytics and its applications [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

## Overview of techniques used in Data Science

Overview of techniques used in Data Science is an official topic in Module 1 of Big Data Concepts. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Overview of techniques used in Data Science using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, overview of techniques used in data science should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

**Topic study guide**

| Aspect | What to prepare |
| --- | --- |
| Definition/identity | What Overview of techniques used in Data Science means in the official course context |
| Study lens | Principle → components or stages → result → application |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 1-[illegible] Overview of techniques used in Data Science [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

**Compare Big Data and Data Warehouse 3 Understanding Introduction to Bigdata : Data –Big data – concepts, characteristics (4 Vs), types of digital data - structured, unstructured and semi structured -Examples of Big data. Data Science/ Big Data Analytics : Definition, Requirement of high performance analytics, benefits, applications and challenges. Techniques used in Data Science - classification, regression and clustering. Data Warehouse: Definition, Coexistence of Big Data and Data Warehouse.**

Compare Big Data and Data Warehouse 3 Understanding Introduction to Bigdata : Data –Big data – concepts, characteristics (4 Vs), types of digital data - structured, unstructured and semi structured -Examples of Big data. Data Science/ Big Data Analytics : Definition, Requirement of high performance analytics, benefits, applications and challenges. Techniques used in Data Science - classification, regression and clustering. Data Warehouse: Definition, Coexistence of Big Data and Data Warehouse. is an official topic in Module 1 of Big Data Concepts. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Compare Big Data and Data Warehouse 3 Understanding Introduction to Bigdata : Data –Big data – concepts, characteristics (4 Vs), types of digital data - structured, unstructured and semi structured -Examples of Big data. Data Science/ Big Data Analytics : Definition, Requirement of high performance analytics, benefits, applications and challenges. Techniques used in Data Science - classification, regression and clustering. Data Warehouse: Definition, Coexistence of Big Data and Data Warehouse. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, compare big data and data warehouse 3 understanding introduction to bigdata : data –big data – concepts, characteristics (4 vs), types of digital data - structured, unstructured and semi structured -examples of big data. data science/ big data analytics : definition, requirement of high performance analytics, benefits, applications and challenges. techniques used in data science - classification, regression and clustering. data warehouse: definition, coexistence of big data and data warehouse. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

**Topic study guide**

| Aspect | What to prepare |
|---|---|
| Definition/identity | What Compare Big Data and Data Warehouse 3 Understanding Introduction to Bigdata : Data –Big data – concepts, characteristics (4 Vs), types of digital data - structured, unstructured and semi structured -Examples of Big data. Data Science/ Big Data Analytics : Definition, Requirement of high performance analytics, benefits, applications and challenges. Techniques used in Data Science - classification, regression and clustering. Data Warehouse: Definition, Coexistence of Big Data and Data Warehouse. means in the official course context |
| Study lens | Principle → components or stages → result → application |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 1-□□ Compare Big Data and Data Warehouse 3 Understanding Introduction to Bigdata : Data –Big data – concepts, characteristics (4 Vs), types of digital data - structured, unstructured and semi structured -Examples of Big data. Data Science/ Big Data Analytics : Definition, Requirement of high performance analytics, benefits, applications and challenges. Techniques used in Data Science - classification, regression and clustering. Data Warehouse: Definition, Coexistence of Big Data and Data Warehouse. □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□□.

### Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
Familiarize the concept of Big Data

  M1.01      Familiarize the concepts of Big Data.                 3
Understanding

  M1.02      Describe Big data Analytics and its applications      3
Understanding

  M1.03      Overview of techniques used in Data Science           3
Understanding

  M1.04      Compare Big Data and Data Warehouse                   3
Understanding
Contents:
Introduction to Bigdata : Data –Big data – concepts, characteristics (4 Vs), types
of digital
data - structured, unstructured and semi structured -Examples of Big data.
Data Science/ Big Data Analytics : Definition, Requirement of high performance
analytics,
benefits, applications and challenges.
Techniques used in Data Science - classification, regression and clustering.
Data Warehouse: Definition, Coexistence of Big Data and Data Warehouse.
```

**Module study routine:** Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 2

# Outline the basic concepts in Big Data Analytics

This module is presented from the official syllabus scope for Big Data Concepts. Use the topic cards for learning and the source transcript for exact coverage.

| As specified | CO-linked |
| --- | --- |
| official hours | see outcomes |

## Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

## Key facts

- Official module title: Outline the basic concepts in Big Data Analytics
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

## Summarize KDD Process.

Summarize KDD Process. is an official topic in Module 2 of Big Data Concepts. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Summarize KDD Process. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, summarize kdd process. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

**Topic study guide**

| Aspect | What to prepare |
|---|---|
| Definition/identity | What Summarize KDD Process. means in the official course context |
| Study lens | Principle → components or stages → result → application |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 2-[illegible] Summarize KDD Process. [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

## Explain data mining techniques

Explain data mining techniques is an official topic in Module 2 of Big Data Concepts. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Explain data mining techniques using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, explain data mining techniques should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

**Topic study guide**

| Aspect | What to prepare |
|---|---|
| Definition/identity | What Explain data mining techniques means in the official course context |
| Study lens | Principle → components or stages → result → application |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 2-□□ Explain data mining techniques □□□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

## Describe data preprocessing methods

Describe data preprocessing methods is an official topic in Module 2 of Big Data Concepts. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Describe data preprocessing methods using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, describe data preprocessing methods should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

**Topic study guide**

| Aspect | What to prepare |
|---|---|
| Definition/identity | What Describe data preprocessing methods means in the official course context |
| Study lens | Principle → components or stages → result → application |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 2-□□ Describe data preprocessing methods □□□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□□.

**Outline data visualization. 3 Understanding Knowledge Discovery in Databases - KDD Process, steps-data cleaning, data integration, data selection, Data Transformation, Data Mining, Pattern Evaluation and Knowledge representation. KDD-advantages and disadvantages. Data Mining-Introduction, Data Mining Goals, Steps to discover knowledge from data (Stages of the Data Mining Process), Overview of data mining Techniques - Association Rules, Classification, Prediction, Clustering, Regression, ANN, Outlier Detection and Genetic Algorithm. Knowledge Representation Methods Data Preprocessing - Need of data preprocessing, data quality, data preprocessing methods- cleaning, integration, transformation, reduction, discretization, normalization. Data Visualization: Definition, Need for Visualization, Advantages.**

Outline data visualization. 3 Understanding Knowledge Discovery in Databases - KDD Process, steps-data cleaning, data integration, data selection, Data Transformation, Data Mining, Pattern Evaluation and Knowledge representation. KDD-advantages and disadvantages. Data Mining- Introduction, Data Mining Goals, Steps to discover knowledge from data (Stages of the Data Mining Process), Overview of data mining Techniques - Association Rules, Classification, Prediction, Clustering, Regression, ANN, Outlier Detection and Genetic Algorithm. Knowledge Representation Methods Data Preprocessing - Need of data preprocessing, data quality, data preprocessing methods- cleaning, integration, transformation, reduction, discretization, normalization. Data Visualization: Definition, Need for Visualization, Advantages. is an official topic in Module 2 of Big Data Concepts. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Outline data visualization. 3 Understanding Knowledge Discovery in Databases - KDD Process, steps-data cleaning, data integration, data selection, Data Transformation, Data Mining, Pattern Evaluation and Knowledge representation. KDD-advantages and disadvantages. Data Mining- Introduction, Data Mining Goals, Steps to discover knowledge from data (Stages of the Data Mining Process), Overview of data mining Techniques - Association Rules, Classification, Prediction, Clustering, Regression, ANN, Outlier Detection and

Genetic Algorithm. Knowledge Representation Methods Data Preprocessing - Need of data preprocessing, data quality, data preprocessing methods- cleaning, integration, transformation, reduction, discretization, normalization. Data Visualization: Definition, Need for Visualization, Advantages. using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, outline data visualization. 3 understanding knowledge discovery in databases - kdd process, steps-data cleaning, data integration, data selection, data transformation, data mining, pattern evaluation and knowledge representation. kdd-advantages and disadvantages. data mining- introduction, data mining goals, steps to discover knowledge from data (stages of the data mining process), overview of data mining techniques - association rules, classification, prediction, clustering, regression, ann, outlier detection and genetic algorithm. knowledge representation methods data preprocessing - need of data preprocessing, data quality, data preprocessing methods- cleaning, integration, transformation, reduction, discretization, normalization. data visualization: definition, need for visualization, advantages. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

**Topic study guide**

| Aspect | What to prepare |
|---|---|
| Definition/identity | What Outline data visualization. 3 Understanding Knowledge Discovery in Databases - KDD Process, steps-data cleaning, data integration, data selection, Data Transformation, Data Mining, Pattern Evaluation and Knowledge representation. KDD-advantages and disadvantages. Data Mining- Introduction, Data Mining Goals, Steps to discover knowledge from data (Stages of the Data Mining Process), Overview of data mining Techniques - Association Rules, Classification, Prediction, Clustering, Regression, ANN, Outlier Detection and Genetic Algorithm. Knowledge Representation Methods Data Preprocessing - Need of data preprocessing, data quality, data preprocessing methods- cleaning, integration, transformation, reduction, discretization, normalization. Data Visualization: Definition, Need for Visualization, Advantages. means in the official course context |
| Study lens | Principle → components or stages → result → application |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 2-□□ Outline data visualization. 3 Understanding Knowledge Discovery in Databases - KDD Process, steps-data cleaning, data integration, data selection, Data Transformation, Data Mining, Pattern Evaluation and Knowledge representation. KDD-advantages and disadvantages. Data Mining-Introduction, Data Mining Goals, Steps to discover knowledge from data (Stages of the Data Mining Process), Overview of data mining Techniques - Association Rules, Classification, Prediction, Clustering, Regression, ANN, Outlier Detection and Genetic Algorithm. Knowledge Representation Methods Data Preprocessing - Need of data preprocessing, data quality, data preprocessing methods- cleaning, integration, transformation, reduction, discretization, normalization. Data Visualization: Definition, Need for Visualization, Advantages. □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

### Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
Outline the basic concepts in Big Data Analytics

  M2.01      Summarize KDD Process.                                3
Understanding

  M2.02      Explain data mining techniques                        5
Understanding

  M2.03      Describe data preprocessing methods                   4
Understanding

  M2.04      Outline data visualization.                           3
Understanding

             Series Test – I                                       1
Contents:
Knowledge Discovery in Databases - KDD Process, steps-data cleaning, data
integration,
data selection, Data Transformation, Data Mining, Pattern Evaluation and
Knowledge
representation. KDD-advantages and disadvantages.

Data Mining- Introduction, Data Mining Goals, Steps to discover knowledge from
data
(Stages of the Data Mining Process), Overview of data mining Techniques -
Association
Rules, Classification, Prediction, Clustering, Regression, ANN, Outlier Detection
```

**Module study routine:** Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 3

# Implement of Exploratory Data Analysis using Python

This module is presented from the official syllabus scope for Big Data Concepts. Use the topic cards for learning and the source transcript for exact coverage.

**As specified**
official hours

**CO-linked**
see outcomes

## Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

## Key facts

- Official module title: Implement of Exploratory Data Analysis using Python
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

## Familiarize the EDA process. 3 Understanding Illustrate step-by-step Exploratory Data

Familiarize the EDA process. 3 Understanding Illustrate step-by-step Exploratory Data is an official topic in Module 3 of Big Data Concepts. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Familiarize the EDA process. 3 Understanding Illustrate step-by-step Exploratory Data using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, familiarize the eda process. 3 understanding illustrate step-by-step exploratory data should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

**Topic study guide**

| Aspect | What to prepare |
|---|---|
| Definition/identity | What Familiarize the EDA process. 3 Understanding Illustrate step-by-step Exploratory Data means in the official course context |
| Study lens | Principle → components or stages → result → application |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 3-□□ Familiarize the EDA process. 3 Understanding Illustrate step-by-step Exploratory Data □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

## 8 Applying Analysis (EDA) using Python.

8 Applying Analysis (EDA) using Python. is an official topic in Module 3 of Big Data Concepts. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify 8 Applying Analysis (EDA) using Python. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, 8 applying analysis (eda) using python. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

**Topic study guide**

| Aspect | What to prepare |
| --- | --- |
| Definition/identity | What 8 Applying Analysis (EDA) using Python. means in the official course context |
| Study lens | Principle → components or stages → result → application |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 3-□□ 8 Applying Analysis (EDA) using Python. □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

## Summarize the concepts of NoSQL

Summarize the concepts of NoSQL is an official topic in Module 3 of Big Data Concepts. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Summarize the concepts of NoSQL using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, summarize the concepts of nosql should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

**Topic study guide**

| Aspect | What to prepare |
|---|---|
| Definition/identity | What Summarize the concepts of NoSQL means in the official course context |
| Study lens | Principle → components or stages → result → application |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 3-[illegible] Summarize the concepts of NoSQL [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

**Outline the concepts of NewSQL 2 Understanding EDA process: Definition, importance, types-univariate, bivariate and multivariate. Exploratory Data Analysis (EDA) using Python: Tools and libraries for EDA (e.g., Python, R, Jupyter Notebooks) Import python libraries - numpy, pandas, matplotlib, seaborn. Read and analyze datasets - reading data, analyzing, checking for duplication, and missing values calculation. Data Reduction, Feature Engineering, univariate analysis (swarmplots and countplots), bivariate analysis (pair plot, violin plot, scatter plot, box plot), multivariate analysis (Correlation Matrix, heat maps). NoSQL: Definition, Types of NoSQL Databases, Use NoSQL - Storing and manipulating data. NewSQL: Characteristics of NewSQL, Comparison of SQL, NoSQL, and NewSQL**

Outline the concepts of NewSQL 2 Understanding EDA process: Definition, importance, types-univariate, bivariate and multivariate. Exploratory Data Analysis (EDA) using Python: Tools and libraries for EDA (e.g., Python, R, Jupyter Notebooks) Import python libraries - numpy, pandas, matplotlib, seaborn. Read and analyze datasets - reading data, analyzing, checking for duplication, and missing values calculation. Data Reduction, Feature Engineering, univariate analysis (swarmplots and countplots), bivariate analysis (pair plot, violin plot, scatter plot, box plot), multivariate analysis (Correlation Matrix, heat maps). NoSQL: Definition, Types of NoSQL Databases, Use NoSQL - Storing and manipulating data. NewSQL: Characteristics of NewSQL, Comparison of SQL, NoSQL, and NewSQL is an official topic in Module 3 of Big Data Concepts. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Outline the concepts of NewSQL 2 Understanding EDA process: Definition, importance, types-univariate, bivariate and multivariate. Exploratory Data Analysis (EDA) using Python: Tools and libraries for EDA (e.g., Python, R, Jupyter Notebooks) Import python libraries - numpy, pandas, matplotlib, seaborn. Read and analyze datasets - reading data, analyzing, checking for duplication, and missing values calculation. Data Reduction, Feature Engineering, univariate analysis (swarmplots and countplots), bivariate analysis (pair plot, violin plot, scatter plot, box plot), multivariate analysis (Correlation Matrix, heat maps).

NoSQL: Definition, Types of NoSQL Databases, Use NoSQL - Storing and manipulating data. NewSQL: Characteristics of NewSQL, Comparison of SQL, NoSQL, and NewSQL using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

## Engineering or professional relevance

In Polytechnic practice, outline the concepts of newsql 2 understanding eda process: definition, importance, types-univariate, bivariate and multivariate. exploratory data analysis (eda) using python: tools and libraries for eda (e.g., python, r, jupyter notebooks) import python libraries - numpy, pandas, matplotlib, seaborn. read and analyze datasets - reading data, analyzing, checking for duplication, and missing values calculation. data reduction, feature engineering, univariate analysis (swarmplots and countplots), bivariate analysis (pair plot, violin plot, scatter plot, box plot), multivariate analysis (correlation matrix, heat maps). nosql: definition, types of nosql databases, use nosql - storing and manipulating data. newsql: characteristics of newsql, comparison of sql, nosql, and newsql should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

**Topic study guide**

| Aspect | What to prepare |
|---|---|
| Definition/identity | What Outline the concepts of NewSQL 2 Understanding EDA process: Definition, importance, types-univariate, bivariate and multivariate. Exploratory Data Analysis (EDA) using Python: Tools and libraries for EDA (e.g., Python, R, Jupyter Notebooks) Import python libraries - numpy, pandas, matplotlib, seaborn. Read and analyze datasets - reading data, analyzing, checking for duplication, and missing values calculation. Data Reduction, Feature Engineering, univariate analysis (swarmplots and countplots), bivariate analysis (pair plot, violin plot, scatter plot, box plot), multivariate analysis (Correlation Matrix, heat maps). NoSQL: Definition, Types of NoSQL Databases, Use NoSQL - Storing and manipulating data. NewSQL: Characteristics of NewSQL, Comparison of SQL, NoSQL, and NewSQL means in the official course context |
| Study lens | Principle → components or stages → result → application |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 3-[illegible] Outline the concepts of NewSQL 2 Understanding EDA process: Definition, importance, types-univariate, bivariate and multivariate. Exploratory Data Analysis (EDA) using Python: Tools and libraries for EDA (e.g., Python, R, Jupyter Notebooks) Import python libraries - numpy, pandas, matplotlib, seaborn. Read and analyze datasets - reading data, analyzing, checking for duplication, and missing values calculation. Data Reduction, Feature Engineering, univariate analysis (swarmplots and countplots), bivariate analysis (pair plot, violin plot, scatter plot, box plot), multivariate analysis (Correlation Matrix, heat maps). NoSQL: Definition, Types of NoSQL Databases, Use NoSQL - Storing and manipulating data. NewSQL: Characteristics of NewSQL, Comparison of SQL, NoSQL, and NewSQL [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

### Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
Implement of Exploratory Data Analysis using Python
  M3.01      Familiarize the EDA process.                          3
Understanding
             Illustrate step-by-step Exploratory Data
  M3.02                                                            8
Applying
             Analysis (EDA) using Python.
  M3.03      Summarize the concepts of NoSQL                       2
Understanding
  M3.04      Outline the concepts of NewSQL                        2
Understanding
Contents:
EDA process: Definition, importance, types-univariate, bivariate and
multivariate.

Exploratory Data Analysis (EDA) using Python: Tools and libraries for EDA (e.g.,
Python, R, Jupyter Notebooks)
Import python libraries - numpy, pandas, matplotlib, seaborn.
Read and analyze datasets - reading data, analyzing, checking for duplication,
and missing
values calculation.
Data Reduction, Feature Engineering, univariate analysis (swarmplots and
countplots),
bivariate analysis (pair plot, violin plot, scatter plot, box plot), multivariate
analysis
(Correlation Matrix, heat maps).
```

**Module study routine:** Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 4

# Demonstrate Map Reduce programming in Hadoop Environment

This module is presented from the official syllabus scope for Big Data Concepts. Use the topic cards for learning and the source transcript for exact coverage.

**As specified**
official hours

**CO-linked**
see outcomes

## Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

## Key facts

- Official module title: Demonstrate Map Reduce programming in Hadoop Environment
- Official duration: As specified
- Official topic entries captured: 6
- The full source excerpt is preserved below for coverage checking.

## Explain the fundamentals of Hadoop

Explain the fundamentals of Hadoop is an official topic in Module 4 of Big Data Concepts. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Explain the fundamentals of Hadoop using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, explain the fundamentals of hadoop should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

**Topic study guide**

| Aspect | What to prepare |
|---|---|
| Definition/identity | What Explain the fundamentals of Hadoop means in the official course context |
| Study lens | Principle → components or stages → result → application |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 4-□□ Explain the fundamentals of Hadoop □□□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

## Concepts of HDFS

Concepts of HDFS is an official topic in Module 4 of Big Data Concepts. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Concepts of HDFS using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, concepts of hdfs should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

**Topic study guide**

| Aspect | What to prepare |
|---|---|
| Definition/identity | What Concepts of HDFS means in the official course context |
| Study lens | Principle → components or stages → result → application |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 4-[illegible] Concepts of HDFS [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

## List the components of Hadoop Ecosystem

List the components of Hadoop Ecosystem is an official topic in Module 4 of Big Data Concepts. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify List the components of Hadoop Ecosystem using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, list the components of hadoop ecosystem should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

**Topic study guide**

| Aspect | What to prepare |
|---|---|
| Definition/identity | What List the components of Hadoop Ecosystem means in the official course context |
| Study lens | Principle → components or stages → result → application |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 4-□□ List the components of Hadoop Ecosystem □□□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

## Illustrate the concept of HDFS and MapReduce

Illustrate the concept of HDFS and MapReduce is an official topic in Module 4 of Big Data Concepts. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Illustrate the concept of HDFS and MapReduce using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, illustrate the concept of hdfs and mapreduce should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

**Topic study guide**

| Aspect | What to prepare |
|---|---|
| Definition/identity | What Illustrate the concept of HDFS and MapReduce means in the official course context |
| Study lens | Principle → components or stages → result → application |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 4-[illegible] Illustrate the concept of HDFS and MapReduce [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

## Use the different function in MapReduce

Use the different function in MapReduce is an official topic in Module 4 of Big Data Concepts. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Use the different function in MapReduce using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, use the different function in mapreduce should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

**Topic study guide**

| Aspect | What to prepare |
|---|---|
| Definition/identity | What Use the different function in MapReduce means in the official course context |
| Study lens | Principle → components or stages → result → application |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 4-□□ Use the different function in MapReduce □□□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□.

**Outline the basic concepts of Hive and Pig 3 Understanding Hadoop: Features of Hadoop, Key Advantages of Hadoop, Hadoop Distributions, Integrated Hadoop Systems Offered by Leading Market Vendors, Cloud-Based Hadoop Solutions Overview of Hadoop Ecosystems and Hadoop Core Components, HDFS, Processing Data with Hadoop Map Reduce Programming: Introduction, Mapper, Reducer, Combiner, Partitioner, Searching, Sorting. Basic concepts of Hive and Pig Text / Reference T/R Book Title/Author T1 T1 Seema Acharya, Subhasini Chellappan, "Big Data Analytics", Wiley, 2015 Jiawei Han,Champaign Micheline Kamber ,Jian Pei Simon Fraser ‘”Data Mining T2 Concepts and Techniques Third Edition” Frank J Ohlhorst, —Big Data and Analytics: Turning Big Data into Big Money, R1 Wiley and SAS Business Series, 2012. R2 Tom White, — Hadoop: The Definitive Guide Third Edition, O‘reily Media, 2012 EMC Education Services, Data Science and Big Data Analytics: Discovering, R3 Analyzing, Visualizing and Presenting Data. John Wiley & Sons, 2015. Publishers – 2012. Online Resources Sl.No Website Link 1 https://hadoop.apache.org/ 2 https://aws.amazon.com/emr/details/hadoop/what-is-hadoop/ 4 https://hive.apache.org/ 5 https://www.tutorialspoint.com/hadoop/index.htm 6 https://pig.apache.org/**

Outline the basic concepts of Hive and Pig 3 Understanding Hadoop: Features of Hadoop, Key Advantages of Hadoop, Hadoop Distributions, Integrated Hadoop Systems Offered by Leading Market Vendors, Cloud-Based Hadoop Solutions Overview of Hadoop Ecosystems and Hadoop Core Components, HDFS, Processing Data with Hadoop Map Reduce Programming: Introduction, Mapper, Reducer, Combiner, Partitioner, Searching, Sorting. Basic concepts of Hive and Pig Text / Reference T/R Book Title/Author T1 T1 Seema Acharya, Subhasini Chellappan, "Big Data Analytics", Wiley, 2015 Jiawei Han,Champaign Micheline Kamber ,Jian Pei Simon Fraser ‘”Data Mining T2 Concepts and Techniques Third Edition” Frank J Ohlhorst, —Big Data and Analytics: Turning Big Data into Big Money, R1 Wiley and SAS Business Series, 2012. R2 Tom White, — Hadoop: The Definitive Guide Third Edition, O‘reily Media, 2012 EMC Education Services, Data Science and Big Data Analytics: Discovering, R3 Analyzing, Visualizing and Presenting Data. John Wiley & Sons, 2015. Publishers – 2012. Online Resources Sl.No Website Link 1 https://hadoop.apache.org/ 2 https://aws.amazon.com/emr/details/hadoop/what-is-hadoop/ 4 https://hive.apache.org/ 5 https://www.tutorialspoint.com/hadoop/index.htm 6 https://pig.apache.org/ is an official topic in Module 4 of Big Data Concepts. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Outline the basic concepts of Hive and Pig 3 Understanding Hadoop: Features of Hadoop, Key Advantages of Hadoop, Hadoop Distributions, Integrated Hadoop Systems Offered by Leading Market Vendors, Cloud-Based Hadoop Solutions Overview of Hadoop Ecosystems and Hadoop Core Components, HDFS, Processing Data with Hadoop Map Reduce Programming: Introduction, Mapper, Reducer, Combiner, Partitioner, Searching, Sorting. Basic concepts of Hive and Pig Text / Reference T/R Book Title/Author T1 T1 Seema Acharya, Subhasini Chellappan, "Big Data Analytics", Wiley, 2015 Jiawei Han,Champaign Micheline Kamber ,Jian Pei Simon Fraser ‘”Data Mining T2 Concepts and Techniques Third Edition” Frank J Ohlhorst, —Big Data and Analytics: Turning Big Data into Big Money, R1 Wiley and SAS Business Series, 2012. R2 Tom White, — Hadoop: The Definitive Guide Third Edition, O‘reily Media, 2012 EMC Education Services, Data Science and Big Data Analytics: Discovering, R3 Analyzing, Visualizing and Presenting Data. John Wiley & Sons, 2015. Publishers – 2012. Online Resources Sl.No Website Link 1 https://hadoop.apache.org/ 2 https://aws.amazon.com/emr/details/hadoop/what-is-hadoop/ 4 https://hive.apache.org/ 5 https://www.tutorialspoint.com/hadoop/index.htm 6 https://pig.apache.org/ using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, outline the basic concepts of hive and pig 3 understanding hadoop: features of hadoop, key advantages of hadoop, hadoop distributions, integrated hadoop systems offered by leading market vendors, cloud-based hadoop solutions overview of hadoop ecosystems and hadoop core components, hdfs, processing data with hadoop map reduce programming: introduction, mapper, reducer, combiner, partitioner, searching, sorting. basic concepts of hive and pig text / reference t/r book title/author t1 t1 seema acharya, subhasini chellappan, "big data analytics", wiley, 2015 jiawei han,champaign micheline kamber ,jian pei simon fraser ‘”data mining t2 concepts and techniques third edition” frank j ohlhorst, —big data and analytics: turning big data into big money, r1 wiley and sas business series, 2012. r2 tom white, — hadoop: the definitive guide third edition, o‘reily media, 2012 emc education services, data science and big data analytics: discovering, r3 analyzing, visualizing and presenting data. john wiley & sons, 2015. publishers – 2012. online resources sl.no website link 1 https://hadoop.apache.org/ 2 https://aws.amazon.com/emr/details/hadoop/what-is-hadoop/ 4 https://hive.apache.org/ 5 https://www.tutorialspoint.com/hadoop/index.htm 6 https://pig.apache.org/ should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

**Topic study guide**

| Aspect | What to prepare |
|---|---|
| Definition/identity | What Outline the basic concepts of Hive and Pig 3 Understanding Hadoop: Features of Hadoop, Key Advantages of Hadoop, Hadoop Distributions, Integrated Hadoop Systems Offered by Leading Market Vendors, Cloud-Based Hadoop Solutions Overview of Hadoop Ecosystems and Hadoop Core Components, HDFS, Processing Data with Hadoop Map Reduce Programming: Introduction, Mapper, Reducer, Combiner, Partitioner, Searching, Sorting. Basic concepts of Hive and Pig Text / Reference T/R Book Title/Author T1 T1 Seema Acharya, Subhasini Chellappan, "Big Data Analytics", Wiley, 2015 Jiawei Han,Champaign Micheline Kamber ,Jian Pei Simon Fraser ‘”Data Mining T2 Concepts and Techniques Third Edition” Frank J Ohlhorst, —Big Data and Analytics: Turning Big Data into Big Money, R1 Wiley and SAS Business Series, 2012. R2 Tom White, —Hadoop: The Definitive Guide Third Edition, O‘reily Media, 2012 EMC Education Services, Data Science and Big Data Analytics: Discovering, R3 Analyzing, Visualizing and Presenting Data. John Wiley & Sons, 2015. Publishers – 2012. Online Resources Sl.No Website Link 1 https://hadoop.apache.org/ 2 https://aws.amazon.com/emr/details/hadoop/what-is-hadoop/ 4 https://hive.apache.org/ 5 https://www.tutorialspoint.com/hadoop/index.htm 6 https://pig.apache.org/ means in the official course context |
| Study lens | Principle → components or stages → result → application |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 4-□□ Outline the basic concepts of Hive and Pig 3 Understanding Hadoop: Features of Hadoop, Key Advantages of Hadoop, Hadoop Distributions, Integrated Hadoop Systems Offered by Leading Market Vendors, Cloud-Based Hadoop Solutions Overview of Hadoop Ecosystems and Hadoop Core Components, HDFS, Processing Data with Hadoop Map Reduce Programming: Introduction, Mapper, Reducer, Combiner, Partitioner, Searching, Sorting. Basic concepts of Hive and Pig Text / Reference T/R Book Title/Author T1 T1 Seema Acharya, Subhasini Chellappan, "Big Data Analytics", Wiley, 2015 Jiawei Han,Champaign Micheline Kamber ,Jian Pei Simon Fraser ‘”Data Mining T2 Concepts and Techniques Third Edition” Frank J Ohlhorst, —Big Data and Analytics: Turning Big Data into Big Money, R1 Wiley and SAS Business Series, 2012. R2 Tom White, —

Hadoop: The Definitive Guide Third Edition, O‘reily Media, 2012 EMC Education Services, Data Science and Big Data Analytics: Discovering, R3 Analyzing, Visualizing and Presenting Data. John Wiley & Sons, 2015. Publishers – 2012. Online Resources Sl.No Website Link 1 https://hadoop.apache.org/ 2 https://aws.amazon.com/emr/details/hadoop/what-is-hadoop/ 4 https://hive.apache.org/ 5 https://www.tutorialspoint.com/hadoop/index.htm 6 https://pig.apache.org/ □□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□□.

## Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
Demonstrate Map Reduce programming in Hadoop Environment

  M4.01      Explain the fundamentals of Hadoop                     2
Understanding
  M4.02      Concepts of HDFS                                       2
Understanding
  M4.03      List the components of Hadoop Ecosystem                2
Understanding
  M4.04      Illustrate the concept of HDFS and MapReduce           4
Applying
  M4.05      Use the different function in MapReduce               3
Applying
  M4.06      Outline the basic concepts of Hive and Pig             3
Understanding
             Series Test -2                                         1
Contents:
Hadoop: Features of Hadoop, Key Advantages of Hadoop, Hadoop Distributions,
Integrated
Hadoop Systems Offered by Leading Market Vendors, Cloud-Based Hadoop Solutions
Overview of Hadoop Ecosystems and Hadoop Core Components, HDFS, Processing
Data with Hadoop
Map Reduce Programming: Introduction, Mapper, Reducer, Combiner, Partitioner,
Searching, Sorting.
Basic concepts of Hive and Pig
Text / Reference

T/R                                         Book Title/Author
```

**Module study routine:** Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

RAPID REFERENCE

## Concept and formula bank

### Answer structure

```
Definition → Principle → Components/steps → Application → Conclusion
```

Use this sequence for descriptive questions when the topic does not require a numerical formula.

### Practical record

```
Aim → Apparatus → Procedure → Observation → Result → Safety
```

Use this sequence for laboratory, workshop and field activities.

RAPID REVISION

## Diagram and source-transcript library

### Module 1 source and topic cards

Jump to official terminology, topic explanations and study guidance.

### Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

### Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

### Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

OFFICIAL SELF-LEARNING

## Activities from the syllabus

Use the extracted official activity wording as the record of required self-learning work.

The official PDF does not expose a separate activities heading in the extracted text; consult the source PDF pages for the exact activity list.

EXAMINATION PREPARATION

## Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

```
The official assessment structure is reproduced from the supplied PDF.
```

**Practice-paper notice:** The model paper below is generated for revision and is not an official examination paper.

ACTIVE RECALL

## Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

### Module 1

**Define or explain Familiarize the concepts of Big Data.. (3 marks)**

**Answer:** Begin with the standard definition or identity of *Familiarize the concepts of Big Data.*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

**Define or explain Describe Big data Analytics and its applications. (3 marks)**

**Answer:** Begin with the standard definition or identity of *Describe Big data Analytics and its applications*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

**Define or explain Overview of techniques used in Data Science. (3 marks)**

**Answer:** Begin with the standard definition or identity of *Overview of techniques used in Data Science*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

**Define or explain Compare Big Data and Data Warehouse 3 Understanding Introduction to Bigdata : Data –Big data – concepts, characteristics (4 Vs), types of digital data - structured, unstructured and semi structured -Examples of Big data. Data Science/ Big Data Analytics : Definition, Requirement of high performance analytics, benefits, applications and challenges. Techniques used**

**in Data Science - classification, regression and clustering. Data Warehouse: Definition, Coexistence of Big Data and Data Warehouse.. (3 marks)**

**Answer:** Begin with the standard definition or identity of *Compare Big Data and Data Warehouse 3 Understanding Introduction to Bigdata : Data –Big data – concepts, characteristics (4 Vs), types of digital data - structured, unstructured and semi structured -Examples of Big data. Data Science/ Big Data Analytics : Definition, Requirement of high performance analytics, benefits, applications and challenges. Techniques used in Data Science - classification, regression and clustering. Data Warehouse: Definition, Coexistence of Big Data and Data Warehouse.*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□ □□□□□□□□□□ □□□□□□□.

## Module 2

**Define or explain Summarize KDD Process.. (3 marks)**

**Answer:** Begin with the standard definition or identity of *Summarize KDD Process.*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□ □□□□□□□□□□ □□□□□□□.

**Define or explain Explain data mining techniques. (3 marks)**

**Answer:** Begin with the standard definition or identity of *Explain data mining techniques*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

**Define or explain Describe data preprocessing methods. (3 marks)**

**Answer:** Begin with the standard definition or identity of *Describe data preprocessing methods*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

**Define or explain Outline data visualization. 3 Understanding Knowledge Discovery in Databases - KDD Process, steps-data cleaning, data integration, data selection, Data Transformation, Data Mining, Pattern Evaluation and Knowledge representation. KDD-advantages and disadvantages. Data Mining-Introduction, Data Mining Goals, Steps to discover knowledge from data (Stages of the Data Mining Process), Overview of data mining Techniques - Association Rules, Classification, Prediction, Clustering, Regression, ANN, Outlier Detection and Genetic Algorithm. Knowledge Representation Methods Data Preprocessing - Need of data preprocessing, data quality, data preprocessing methods- cleaning, integration, transformation, reduction, discretization, normalization. Data Visualization: Definition, Need for Visualization, Advantages.. (3 marks)**

**Answer:** Begin with the standard definition or identity of *Outline data visualization. 3 Understanding Knowledge Discovery in Databases - KDD Process, steps-data cleaning, data integration, data selection, Data Transformation, Data Mining, Pattern Evaluation and Knowledge representation. KDD-advantages and disadvantages. Data Mining-Introduction, Data Mining Goals, Steps to discover knowledge from data (Stages of the Data Mining Process), Overview of data mining Techniques - Association Rules, Classification, Prediction, Clustering, Regression, ANN, Outlier Detection and Genetic Algorithm. Knowledge Representation Methods Data Preprocessing - Need of data preprocessing, data quality, data preprocessing methods- cleaning, integration, transformation, reduction, discretization, normalization. Data Visualization: Definition, Need for Visualization, Advantages.*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

## Module 3

**Define or explain Familiarize the EDA process. 3 Understanding Illustrate step-by-step Exploratory Data. (3 marks)**

**Answer:** Begin with the standard definition or identity of *Familiarize the EDA process. 3 Understanding Illustrate step-by-step Exploratory Data*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

**Define or explain 8 Applying Analysis (EDA) using Python.. (3 marks)**

**Answer:** Begin with the standard definition or identity of *8 Applying Analysis (EDA) using Python.*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

**Define or explain Summarize the concepts of NoSQL. (3 marks)**

**Answer:** Begin with the standard definition or identity of *Summarize the concepts of NoSQL*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

**Define or explain Outline the concepts of NewSQL 2 Understanding EDA process: Definition, importance, types-univariate, bivariate and multivariate. Exploratory Data Analysis (EDA) using Python: Tools and libraries for EDA (e.g., Python, R, Jupyter Notebooks) Import python libraries - numpy, pandas, matplotlib, seaborn. Read and analyze datasets - reading data, analyzing, checking for duplication, and missing values calculation. Data Reduction, Feature Engineering, univariate analysis (swarmplots and countplots), bivariate analysis (pair plot, violin plot, scatter plot, box plot), multivariate analysis (Correlation Matrix, heat maps). NoSQL: Definition, Types of NoSQL Databases, Use NoSQL - Storing and manipulating data. NewSQL: Characteristics of NewSQL, Comparison of SQL, NoSQL, and NewSQL. (3 marks)**

**Answer:** Begin with the standard definition or identity of *Outline the concepts of NewSQL 2 Understanding EDA process: Definition, importance, types-univariate, bivariate and multivariate. Exploratory Data Analysis (EDA) using Python: Tools and libraries for EDA (e.g., Python, R, Jupyter Notebooks) Import python libraries - numpy, pandas, matplotlib, seaborn. Read and analyze datasets - reading data, analyzing, checking for duplication, and missing values calculation. Data Reduction, Feature Engineering, univariate analysis (swarmplots and countplots), bivariate analysis (pair plot, violin plot, scatter plot, box plot), multivariate analysis (Correlation Matrix, heat maps). NoSQL: Definition, Types of NoSQL Databases, Use NoSQL - Storing and manipulating data. NewSQL: Characteristics of NewSQL, Comparison of SQL, NoSQL, and NewSQL*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

## Module 4

**Define or explain Explain the fundamentals of Hadoop. (3 marks)**

**Answer:** Begin with the standard definition or identity of *Explain the fundamentals of Hadoop*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

**Define or explain Concepts of HDFS. (3 marks)**

**Answer:** Begin with the standard definition or identity of *Concepts of HDFS*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

**Define or explain List the components of Hadoop Ecosystem. (3 marks)**

**Answer:** Begin with the standard definition or identity of *List the components of Hadoop Ecosystem*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

**Define or explain Illustrate the concept of HDFS and MapReduce. (3 marks)**

**Answer:** Begin with the standard definition or identity of *Illustrate the concept of HDFS and MapReduce*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

PRACTICE ONLY

# Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

### Module 1

1-mark definition: State the meaning of **Familiarize the concepts of Big Data.**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

### Module 2

1-mark definition: State the meaning of **Summarize KDD Process.**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

### Module 3

1-mark definition: State the meaning of **Familiarize the EDA process. 3 Understanding Illustrate step-by-step Exploratory Data**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

### Module 4

1-mark definition: State the meaning of **Explain the fundamentals of Hadoop**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

QUICK REVISION

## Exam checklist

### Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

### Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

REFERENCE VOCABULARY

## Glossary

**Study glossary**

| Term | Meaning in this lesson |
|---|---|
| Course Outcome | A capability the student should demonstrate after completing the course. |
| Module | An official syllabus unit with defined topics and hours. |
| CIE | Continuous Internal Evaluation. |
| ESE | End Semester Examination. |
| Source transcript | A preserved extract of the official syllabus used for coverage checking. |

SOURCE-SUPPORTED REFERENCES

# References and web resources

## References listed in the official PDF

References are included in the official source PDF.

## Web resources listed in the official PDF

- Sl.No Website Link https://hadoop.apache.org/
- https://aws.amazon.com/emr/details/hadoop/what-is-hadoop/ https://hive.apache.org/
- https://www.tutorialspoint.com/hadoop/index.htm https://pig.apache.org/

IN-PAGE SEARCH

# Search this lesson